

Model Selection

Post-Inference Problem

1

In previous classes

You learned

- how to select a model using *stepwise* algorithms and *regularized* models
- how to use an estimated model to make inference about the population

Can we use the model selected by stepwise or regularized methods to make inference?

2

The post-inference problem



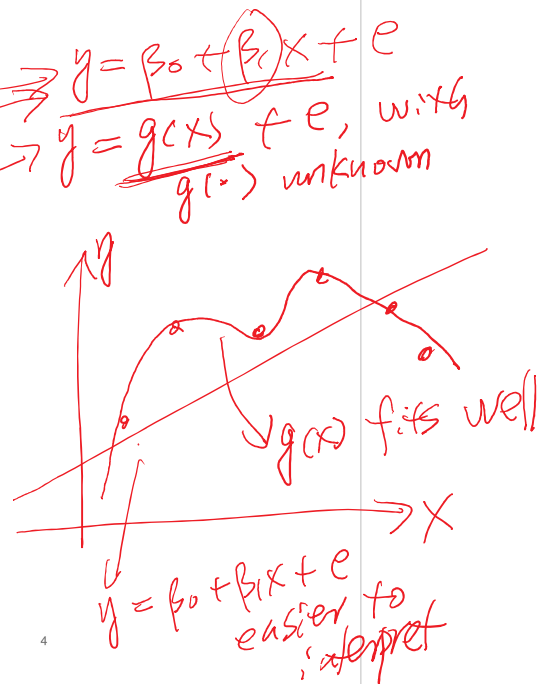
3

Beyond variable selection, there are many aspects involved in selecting a model and a method to estimate it, for example,

- Do we want a parametric or non-parametric approach?
- Do we want to assume a functional form for the relationship between Y and $\mathbf{X}$ (e.g., linear, quadratic, exponential)?
- Prediction vs inference
- Is model interpretability important?
- Which variables should be included in the model?

Taylor expansion.

$$g(x) = x_0 + g'(x_0)(x - x_0) + \frac{g''(x_0)}{2!}(x - x_0)^2 + \dots$$



Simulation Study

Q: If we use data to select a model, can we fit the selected model using the same data to make inference??

To examine this problem, we'll use a simulation study so that

- you know the values of parameters you are trying to infer
- you can quantify errors of your inferential procedure due to randomness
- you can repeat the same scenario multiple times to quantify error rates

How to prove it?
- Math proof is hard.
- Simulation $n \rightarrow \infty$

7/100

Simulation Design

Scroll down to see full content

1. Setting (variables): a response variable Y and $p = 10$ covariates with Normal distribution

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + e$$

$e \sim N(0, \sigma^2)$

2. Setting (coefficients): set all coefficients to 0, so none of the generated covariates will have an effect on Y

$$y = \beta_0 + e \quad \beta_0 = 2, \quad \sigma^2 = 1$$

we expect the intercept only model to be better than any LR that includes any of the covariates

$\rightarrow \beta_1 = \beta_2 = \dots = \beta_{10} = 0$
(imply that none of the predictors should be selected)

3. Generate 100 observations (n) for this setting
4. Apply the forward selection algorithm to select at most 3 variables among the 10 available, using the adj R^2

$$N(2, 1)$$

5. Use the selected model and the same dataset to make inference about the population

← bad idea

6. Replicate this study 1,000 times (rep) and compute the type I error.



Generating data

In the worksheet, we included a function that generates all the data **at once**!

`data` is a tibble with 100000 (100 x 1000) rows and 12 (10 covariates + 1 response + 1 rep ID) columns

- `data` should be treated as 1000 datasets of size 100
- the variable `replicate` identifies each dataset

7

```
1 dim(dataset)
[1] 100000 12
1 length(unique(dataset$replicate))
[1] 1000
```

```
1 head(dataset,2) %>% knitr::kable()
```

X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	Y	replicate
8.07	8.86	15.62	3.37	2.79	0.77	5.65	-8.65	7.52	9.39	-6.99	1
-3.96	3.59	15.84	3.05	1.38	-0.01	-14.54	7.11	-9.83	-3.03	21.06	1

```
1 tail(dataset,2) %>% knitr::kable()
```

X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	Y	replicate
6.97	15.02	6.46	-3.43	13.39	6.20	-14.12	-11.21	15.91	15.14	10.76	1000
-6.55	-1.01	6.46	0.97	-0.33	2.86	10.33	-17.67	4.20	2.07	0.28	1000

Selection Function

We created a function for you:

- takes a dataset as an input
- runs the forward selection algorithm and selects at most 3 variables using $\text{adj}R^2$
- fits LS on the selected variables

9

```

1 forward_selection_function <- function(dataset){
2   sel_model <- regsubsets(x = Y ~ .,
3                           nvmax = 3,
4                           data = dataset,
5                           method = "forward",
6   )
7
8   sel_model_summary <- summary(sel_model)
9
10  adj_r2_min = which.max(sel_model_summary$adjr2)
11
12  selected_var <- names(coef(sel_model, adj_r2_min))[-1]
13
14  data_subset <- dataset %>%
15    select(all_of(selected_var), Y)
16
17  selected_model <- lm(Y ~ ., data = data_subset)
18

```

10

For example

Using the selection function in the first dataset

```
1 selected_model_LS <- forward_selection_function(full_models$data[[1]])
2
3 selected_model_LS%>%
4   tidy()%>%
5   mutate_if(is.numeric, round, 3) %>%
6   knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	4.325	1.354	3.194	0.002
X4	0.154	0.106	1.456	0.149
X10	-0.110	0.106	-1.039	0.301

`full_models` has 3 objects, one is a list of datasets. You will use `map` to pass this function to the whole list!!

The test

We can use the test in `glance` to compare the selected model versus the intercept-only model

```
1 glance(selected_model_LS) %>%  
2   pull(p.value) %>%  
3   round(4)
```

value

0.2427

In this case, we fail to reject the null hypothesis (as expected!)

Recall that all coefficients are 0!

You will use `map` to pass this function to all selected models!!

p-values

	replicate	data	fs_model	F_pvalue
	<int>	<list>	<list>	<dbl>
X_1	1	1 <tibble [100 × 11]>	<lm>	0.243
X_2	2	2 <tibble [100 × 11]>	<lm>	0.138
X_3	3	3 <tibble [100 × 11]>	<lm>	0.0320
X_4	4	4 <tibble [100 × 11]>	<lm>	0.314
X_5	5	5 <tibble [100 × 11]>	<lm>	0.142
X_6	6	6 <tibble [100 × 11]>	<lm>	0.106
X_7	7	7 <tibble [100 × 11]>	<lm>	0.0701
X_8	8	8 <tibble [100 × 11]>	<lm>	0.00299
X_9	9	9 <tibble [100 × 11]>	<lm>	0.0730
X_{10}	10	10 <tibble [100 × 11]>	<lm>	0.00823

Handwritten notes:
 X_3 is significant (~~Fail to~~ Reject $H_0: \beta_3 = 0$)
 X_8 is significant (Reject $H_0: \beta_8 = 0$)
 X_{10}

Wait! In some datasets there is enough evidence to reject the null hypothesis??





A post-inference problem

The problem : we are using the same data to find the variables and fit the model

In the sample at hand, the adjusted R^2 improved after selecting these variables. Thus, we have a much higher chance of wrongly rejecting H_0

A possible solution: split the dataset into two parts: one to select the model and the other one for inference.

Would that solve the problem?

17

Splitting the data

To implement this solution, this time we are going to split each dataset into two parts:

- one part will be used to select a model, and
- the other part for inference.

since the sample size has changed, for a fair comparison, we'll fit the selected models and test them on the first half, then recompute the error rate.

You will use the functions map, map_db1, map2 and update. Let's take a look at them.

18

The `map` function

We can use the function `map` to efficiently apply the same function to all datasets at once (although a loop is run behind scenes)!

- the function `map` applies a function to the list of datasets
- the output of `map` is also a list

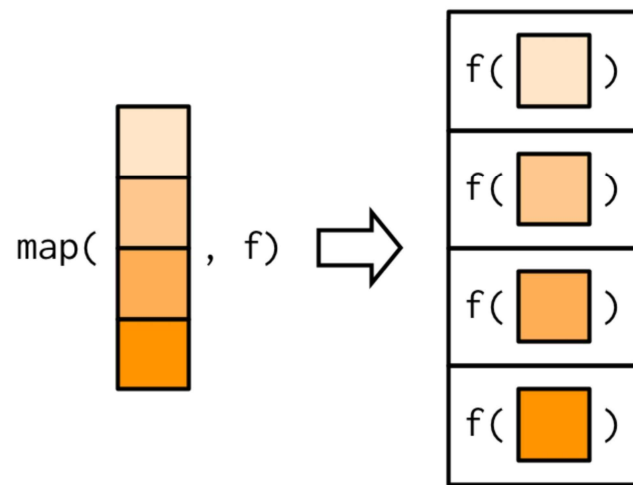


Image from [Advanced R](#), by H. Wickham

For example (from [Advanced R: 9.2](#))

```
1 triple <- function(x) x * 3
2
3 map(1:3, triple)
```

```
[[1]]
[1] 3
```

```
[[2]]
[1] 6
```

```
[[3]]
[1] 9
```

Note that the output of `map` is a list!

Special `map` functions

The `map` function is very general because any type of object can be stored in a list

To return a list with a simpler structure we can use its variants:

`map_lgl()`, `map_int()`, `map_dbl()`, and `map_chr()`

For example (from [Advanced R: 9.2.1](#)), `map_dbl()` always returns a double vector

```
1 map_dbl(mtcars, mean)
```

mpg	cyl	disp	hp	drat	wt	qsec
20.090625	6.187500	230.721875	146.687500	3.596563	3.217250	17.848750
vs	am	gear	carb			
0.437500	0.406250	3.687500	2.812500			

Advanced R: 9.2.2. Anonymous functions

You can write your own function directly in the second argument

```
1 map_dbl(mtcars, function(x) length(unique(x)))
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
25	3	27	22	22	29	30	2	2	3	6

for loop in
R is slow

Advanced R: 9.2.2. Shortcuts

The shortcut `~` can be used to write only the body of the function and the first argument (`.x`) will be passed to it

```
1 map_dbl(mtcars, ~ length(unique(.x)))
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
25	3	27	22	22	29	30	2	2	3	6

23

Example

```
1 mtcars_list <- list(mtcars, mtcars)
2 map(mtcars_list, ~ lm(mpg ~ ., data = .x))
```

[[1]] *iteration 1*

Call:

lm(formula = mpg ~ ., data = .x)

Coefficients:

(Intercept)	cyl	disp	hp	drat	wt
12.30337	-0.11144	0.01334	-0.02148	0.78711	-3.71530
qsec	vs	am	gear	carb	
0.82104	0.31776	2.52023	0.65541	-0.19942	

[[2]] *iteration 2*

Call:

lm(formula = mpg ~ ., data = .x)

Coefficients:

(Intercept)	cyl	disp	hp	drat	wt
12.30337	-0.11144	0.01334	-0.02148	0.78711	-3.71530
qsec	vs	am	gear	carb	
0.82104	0.31776	2.52023	0.65541	-0.19942	

24

For example (from your worksheet)

```
1 full_models <-  
2   dataset %>%  
3   group_by(replicate) %>%  
4   nest() %>%  
5   mutate(models = map(.x = data, ~ lm(Y ~ ., data = .x)))`
```

- a list of datasets is created with `group_by()` and `nest`
- `map` is used to fit LS to each dataset: the function `lm` (f in the figure) is applied to each element of `data`

the first argument `data` can be written as `.x = data` (more useful when there are 2 arguments, see `map2`)

- the output is a list containing the elements `replicate`, `data`, and `models` (the fitted models)!!

25

map2

It is the same as `map` but takes 2 arguments

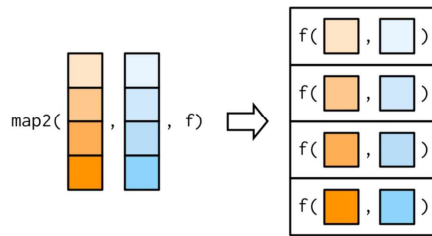


Image from 9.4.2: [Advanced R](#), by H. Wickham

From the worksheet:

```
1 ls_fit = map2(.x = data, .y = lasso_selected_covariates,  
2              ~lm(Y ~ ., data = .x[,c(.y, 'Y')]))
```

update

```
1 # Fit an initial linear model
2 model1 <- lm(mpg ~ wt + hp, data = mtcars)
3
4 # Update the model to add another predictor (qsec)
5 model2 <- update(model1, . ~ . + qsec)
6
7 # Print the updated model
8 summary(model2)
```

Call:

```
lm(formula = mpg ~ wt + hp + qsec, data = mtcars)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.8591	-1.6418	-0.4636	1.1940	5.6092

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.61053	8.41993	3.279	0.00278 **
wt	-4.35880	0.75270	-5.791	3.22e-06 ***
hp	-0.01782	0.01498	-1.190	0.24418
qsec	0.51082	0.12022	4.162	0.000162 ***

27

**Work on section 2 of
worksheet_07**

28

Summary: post-inference problem

Scroll down to see full content

- we can not use the same data to select variables of the model and to conduct inference (“doble dipping”).
- the inference results given by the `lm` are not valid (as seen in the first part of the worksheet).
- if we split the data, we can use one part to select and the other part to estimate and build tests.
- more sophisticated methods have been proposed to address this problem (beyond the scope of this course).

29